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$$\begin{aligned} I &= \int \frac{\ln(\cos x)}{\cos^2 x} dx \\ &= \int \ln(\cos x) \cdot \frac{1}{\cos^2 x} dx \\ \begin{cases} u = \ln(\cos x) \\ dv = \frac{1}{\cos^2 x} dx \end{cases} &\Rightarrow \begin{cases} du = -\tan x dx \\ v = \tan x \end{cases} \\ I &= \ln(\cos x) \tan x - \int \tan x (-\tan x) dx \\ &= \ln(\cos x) \tan x + \int \tan^2 x dx \\ &= \ln(\cos x) \tan x + \int \sec^2 x - 1 dx \\ &= \ln(\cos x) \tan x + \int \sec^2 x dx - \int dx \\ &= \ln(\cos x) \tan x + \tan x - x + c \end{aligned}$$

References